

G3 — Progress Report

Report 01 · Phase A — Scoping & Design

AMS Intake Platform · Algorithmic Intelligence Engine

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Status: Work in progress

Repository: <https://github.com/hodncgi/AMSCatBot>

1. Purpose of This Report

This document is progress report 01 of 8. It focuses on Phase A — Scoping & Design. We use these reports to show incremental delivery to management without waiting for the full project to be complete.

We align with Group 2 on API contracts, draft NBQ and Change Risk specifications, and validate our local AI stack choice (LM Studio + Mistral 7B Instruct).

2. Phase Objectives

- Define JSON API contracts for Group 2 integration.
- Document NBQ and Change Risk business rules for the POC.
- Compare local AI options and select LM Studio + Mistral.
- Produce an architecture view of the G3 engine.

3. Work Performed

- Workshop with Group 2: mandatory fields and response shapes.
- Manual drafting of algorithm specifications.
- Comparative analysis: LM Studio vs Ollama vs rules-only fallback.
- Validated choice: LM Studio + Mistral 7B Instruct on port 1234.

4. Deliverables

- Specification notes for NBQ and Change Risk.
- Architecture diagram and data-flow description.
- Technology stack decision record.

5. How to Verify

The following checks allow management or Group 2 to confirm this phase's outcomes:

- Review API contract draft with Group 2 referent.
- Confirm Mistral model choice and local inference constraints.

Cumulative Project Status

The table below shows where we stand across all implementation phases. Earlier phases are marked completed; this report focuses on the current phase.

Phase	Status
Phase A — Scoping & Design	In progress — focus of this report
Phase B — Backend & NBQ Algorithm	Planned
Phase C — Change Risk Algorithm & Test	Planned

Bench	
Phase D — CGI Knowledge Base	Planned
Phase E — RAG & AI Service	Planned
Phase F — Chatbot Interface (Figma → React)	Planned
Phase G — Testing, Hardening & G2 Integration	Planned
Phase H — Finalization & Delivery	Planned

6. Next Phase Preview

Once Phase A is validated, we move to Phase B — Backend & NBQ Algorithm:

We stand up the FastAPI backend and implement the Next-Best Question algorithm with a Mistral-generated justification field.

7. Reminder — Group 3 Scope

We deliver the NBQ algorithm, Change Risk scoring (decision-support with auditable drivers), the onboarding chatbot (RAG + Mistral via LM Studio), and the documented FastAPI API. Final business sign-off on change risk remains with Group 2 and AMS leads.

Code repository: <https://github.com/hodncgi/AMSChatBot>

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Group 3 — Algorithmic Intelligence Engine